

18.6 Linear Algebra Lecture 4-6

Textbook Problem Sets - Sections 2.6, 2.7, 3.1

• Problem Set 2.6

$$) \rightarrow \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} x = b \Rightarrow \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} x = c$$

$$\begin{array}{l} x+y=5 \\ x+2y=7 \end{array} \rightarrow \begin{array}{l} x+y=5 \\ y=2 \end{array} \quad \begin{bmatrix} 1 & 1 & 5 \\ 1 & 2 & 7 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 5 \\ 0 & 1 & 2 \end{bmatrix}$$

l_{21} = Subtract 1 times row 1 from row 2.

$$L = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \quad l_{21} = 1 \text{ multiplied row 1} \quad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$L = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} x_1 = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

$$L(Ux = c) \rightarrow LUx = Lc \rightarrow Ax = b$$

$$A = LU$$

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} x_1 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} x_1 = \begin{bmatrix} 7 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} x_1 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} x_1 = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$$

$$\Rightarrow L(Ux = c) \Rightarrow Ax = b$$

$$(LUx = Lc) \Rightarrow Ax = b$$

$$3) (Ax = b) \rightarrow (Ux = c)$$

↳ Forward Elimination.

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 6 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \\ 11 \end{bmatrix} \quad \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 2 & 5 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \\ 6 \end{bmatrix} \quad \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \\ 2 \end{bmatrix}$$

↳ $l_{31} = 1$ multiplied
 $l_{32} = 2$

$$l_{31} = 1$$

$$l_{32} = 2$$

$$\rightarrow L = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & -2 & 1 \end{bmatrix}$$

Row 3 of $[A \ b] = (-1 \cdot \text{Row 1} - 2 \cdot \text{Row 2} + 1 \cdot \text{Row 3})$
of $[U \ c]$

↳ ~~$[1 \ 3 \ 6]$ of $[A \ b] = (1 \times [x \ y \ z])$~~

$$[A \ b] = (1 \times (x + y + z = 5) + 2(y + 2z = 2) + 1 \times (z = 2))$$

$$= (x + 3y + 6z = 11)$$

$$5) EA = U?$$

$$E^{-1} = L$$

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix}$$

~~$L = E^{-1}$~~ $EA = U = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 4 & 2 \\ 0 & 0 & 5 \end{bmatrix}$

$$L = E^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix}$$

$$LU = A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 4 & 2 \\ 6 & 3 & 5 \end{bmatrix} \quad \checkmark$$

$$7) A = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 2 & 2 \\ 3 & 4 & 5 \end{bmatrix} \xrightarrow{00} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 3 & 4 & 5 \end{bmatrix} \xrightarrow{00} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 4 & 2 \end{bmatrix} U = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix} E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix}$$

$$\cancel{E_{32} E_{31} E_{21} U} = \cancel{E_{32} E_{31}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$E_{21}^{-1} E_{31}^{-1} E_{32}^{-1} U = E_{21}^{-1} E_{31}^{-1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & +2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} = E_{21}^{-1} E_{31}^{-1} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 4 & 2 \end{bmatrix}$$

$$\cancel{E_{21}^{-1} E_{31}^{-1}} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 4 & 2 \end{bmatrix} = E_{21}^{-1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ +3 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 4 & 2 \end{bmatrix} = E_{21}^{-1} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 3 & 4 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ +2 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 3 & 4 & 5 \end{bmatrix} = A = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 2 & 2 \\ 3 & 4 & 5 \end{bmatrix}$$

$$00 L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}$$

$$9) \begin{bmatrix} 0 & 2 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} d & e \\ 0 & f \end{bmatrix} \quad \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 2 \\ 1 & 2 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 \\ e & 1 \\ m & n & 1 \end{bmatrix} \begin{bmatrix} d & e & g \\ 0 & f & h \\ 0 & 0 & i \end{bmatrix}$$

1d + 0e =

$$1d + 0e = d \quad 0$$

$$1e + 0f = e \quad 0$$

$$ed + (1)(0) = 0$$

gives

$$1e + 1f = f$$

$$\therefore e = 0$$

$$\therefore ed = 0$$

not

$$1d + 0e = 0$$

$$ed = 0$$

d = 0 not allowed

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 2 \\ 1 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ e & 1 & 0 \\ m & n & 1 \end{bmatrix} \begin{bmatrix} d & e & g \\ 0 & f & h \\ 0 & 0 & i \end{bmatrix}$$

$$\therefore 1d + 0 + 0 = 1 \Rightarrow d = 1$$

$$1e = 1 \Rightarrow e = 1$$

$$ed = 1 \Rightarrow e = 1$$

11) $D \rightarrow$ diagonal pivot matrix

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad D = \begin{bmatrix} 2 & & 0 \\ & 3 & \\ 0 & & 7 \end{bmatrix} \checkmark$$

$A = LU :$

$$U = \begin{bmatrix} 2 & 4 & 8 \\ 0 & 3 & 9 \\ 0 & 0 & 7 \end{bmatrix}$$

$A = LDU :$

$$U = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \checkmark$$

13) $A = \begin{bmatrix} a & a & a & a \\ a & b & b & b \\ a & b & c & c \\ a & b & c & d \end{bmatrix} \quad L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ +1 & 1 & 0 & 0 \\ 0 & +1 & 1 & 0 \\ 0 & 0 & +1 & 1 \end{bmatrix} \checkmark$

$$U = \begin{bmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & 0 & c-b & c-b \\ 0 & 0 & 0 & d-c \end{bmatrix}$$

∞

$a \neq 0$

$b-a \neq 0$

$c-b \neq 0$

$d-c \neq 0$

$\begin{cases} b \neq a \\ c \neq b \\ d \neq c \end{cases} \text{ conditions} \checkmark$

15) $L = \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix} \quad U = \begin{bmatrix} 2 & 4 \\ 0 & 1 \end{bmatrix} \quad b = \begin{bmatrix} 2 \\ 11 \end{bmatrix} \quad Lc = b$

$E = L^{-1} = \begin{bmatrix} 1 & 0 \\ -4 & 1 \end{bmatrix}$

$L^{-1}b = \begin{bmatrix} 1 & 0 \\ -4 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 11 \end{bmatrix} = c$
 $2 \times 2 \quad 2 \times 1$

$c = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \checkmark$

~~$L^{-1}b = c \quad c = bL^{-1}$
 $c = \begin{bmatrix} 1 & 0 \\ -4 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 11 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \quad \begin{bmatrix} 2 \\ 11 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -4 & 1 \end{bmatrix}$
 $= \begin{bmatrix} 2 & 0 \\ -4 & 11 \end{bmatrix}$~~

$Ux = c$

$x = U^{-1}c =$

$U^{-1} = \begin{bmatrix} 2 & 4 & | & 1 & 0 \\ 0 & 1 & | & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & | & 1/2 & -2 \\ 0 & 1 & | & 0 & 1 \end{bmatrix}$
 $\begin{matrix} -40 & -2 \\ +51 & 17 \\ & 51 \end{matrix}$

$x = \begin{bmatrix} 1/2 & -2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} -5 \\ 3 \end{bmatrix} \checkmark$
 $2 \times 2 \quad 2 \times 1$

$A = LU = \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} 2 & 4 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 8 & 17 \end{bmatrix}$

$Ax = b \quad Ax = \begin{bmatrix} 2 & 4 \\ 8 & 17 \end{bmatrix} \begin{bmatrix} -5 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 11 \end{bmatrix} \checkmark$

17) a) $L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$ L goes to I

b) $I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ -l_{21} & 1 & 0 \\ -l_{31} & -l_{32} & 1 \end{bmatrix}$ I goes to L^{-1}
Specifically $\Rightarrow$ multipliers are reversed.

c) $LU = ?$

LU goes to U

$LU L^{-1} = U$

19) $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}$

$A = LU$: $L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ $U = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ $\rightarrow +1$ as E had -1

$A = LDL^T$: $L = \begin{bmatrix} 1 & 0 & 0 \\ +1 & 1 & 0 \\ 0 & +1 & 1 \end{bmatrix}$ $D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ $L^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & +1 & 1 \end{bmatrix}$

$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & +1 & 1 \end{bmatrix}$

• Section 2.7 Problem Set

1) $A = \begin{bmatrix} 1 & 0 \\ 9 & 3 \end{bmatrix} \therefore A^T = \begin{bmatrix} 1 & 9 \\ 0 & 3 \end{bmatrix}$ ✓

$A^{-1} \Rightarrow \left[\begin{array}{cc|cc} 1 & 0 & 1 & 0 \\ 9 & 3 & 0 & 1 \end{array} \right] \Rightarrow \left[\begin{array}{cc|cc} 1 & 0 & 1 & 0 \\ 0 & 3 & -9 & 1 \end{array} \right] \Rightarrow \left[\begin{array}{cc|cc} 1 & 0 & 1 & 0 \\ 0 & 1 & -3 & 1/3 \end{array} \right]$

$\therefore A^{-1} = \begin{bmatrix} 1 & 0 \\ -3 & 1/3 \end{bmatrix}$ ✓

$(A^{-1})^T = \begin{bmatrix} 1 & 0 \\ -3 & 1/3 \end{bmatrix}^T = \begin{bmatrix} 1 & -3 \\ 0 & 1/3 \end{bmatrix}$ ✓

$(A^T)^{-1} \Rightarrow \left[\begin{array}{cc|cc} 1 & 9 & 1 & 0 \\ 0 & 3 & 0 & 1 \end{array} \right] \Rightarrow \left[\begin{array}{cc|cc} 1 & 0 & 1 & -3 \\ 0 & 3 & 0 & 1 \end{array} \right]$

$\Rightarrow \left[\begin{array}{cc|cc} 1 & 0 & 1 & -3 \\ 0 & 1 & 0 & 1/3 \end{array} \right]$

$(A^T)^{-1} = \begin{bmatrix} 1 & -3 \\ 0 & 1/3 \end{bmatrix}$ ✓

$(A^{-1})^T = (A^T)^{-1} = \begin{bmatrix} 1 & -3 \\ 0 & 1/3 \end{bmatrix}$ ✓

3) a) $(AB)^{-1})^T \quad (A^{-1})^T (B^{-1})^T$
 $(AB)^{-1})^T = (B^{-1})^T (A^{-1})^T$

b) $U \rightarrow$ upper triangular $\begin{bmatrix} \nabla & \\ 0 & \nabla \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ ✓

$(U^{-1})^T = (U^T)^{-1} \rightarrow$ lower triangular. ✓

5a) $x^T A y = \underbrace{\begin{bmatrix} 0 & 1 \end{bmatrix}}_{1 \times 2} \underbrace{\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}}_{2 \times 3} \underbrace{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}}_{3 \times 1} = \begin{bmatrix} 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = 5$ ✓

b) $x^T A = \begin{bmatrix} 4 & 5 & 6 \end{bmatrix}$ ✓

c) $A y = \underbrace{\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}}_{2 \times 3} \underbrace{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}}_{3 \times 1} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$ ✓

7 a) True x Only $A = A^T$ makes it symmetric

b) False ✓

c) True ✓

d) True ✓

$$9) \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$P_1 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \quad P_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \quad P_1 P_2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \quad P_1 P_2 \neq P_2 P_1$$

$$P_2 P_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$11) A = \begin{bmatrix} 0 & 0 & 6 \\ 1 & 2 & 3 \\ 0 & 4 & 5 \end{bmatrix}$$

$P A \rightarrow$ upper triangular?

$$\hookrightarrow P_u = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$P_u H = \begin{bmatrix} 0 & 4 & 5 \\ 1 & 2 & 3 \\ 0 & 0 & 6 \end{bmatrix}$$

$$P A P_2 = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix} P_2 \text{ or } P_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 3 & 2 & 1 \\ 5 & 4 & 0 \\ 6 & 0 & 0 \end{bmatrix} \quad \times$$

$\hookrightarrow P_1$ required to be diff from P_u

$$13) a) I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} P \rightarrow \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \neq \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$P^3 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}^3 = I \quad \checkmark$$

$$b) \hat{P} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \hat{P}^4 \neq I$$

$$15) a.) P^T \text{ sends Row } 1^{\times 4} \text{ to Row } 4^{\times 1}$$

$$b.) P = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} = P^T$$

$$17) a) S = \begin{bmatrix} 3 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{or } \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$b) S = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \times \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$$

$$c) S = \begin{bmatrix} -3 & 1 \\ 1 & -2 \end{bmatrix} \times \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

$$21) S = \begin{bmatrix} 2 & 4 & 8 \\ 4 & 3 & 9 \\ 8 & 9 & 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 2 & 4 & 8 \\ 0 & -5 & -7 \\ 0 & -7 & -16 \end{bmatrix} \quad [2 \times 2] = \begin{bmatrix} -5 & -7 \\ -7 & -16 \end{bmatrix} \rightarrow 32$$

$$S = \begin{bmatrix} 1 & b & c \\ b & d & e \\ c & e & f \end{bmatrix} \quad \therefore \begin{bmatrix} 1 & b & c \\ 0 & d-b^2 & e-bc \\ 0 & e-b^2 & f-bc \end{bmatrix} \quad [2 \times 2] = \begin{bmatrix} d-b^2 & e-bc \\ e-b^2 & f-bc \end{bmatrix} \quad \times$$

$$23) A = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad P = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix} \quad A = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

~~scribbled out text~~

$$U = L - I$$

$$24) A = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 3 & 8 \\ 2 & 1 & 1 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 & 2 \\ 0 & 3 & 8 \\ 2 & 1 & 1 \end{bmatrix} = LU$$

$$P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 & 2 \\ 0 & 3 & 8 \\ 0 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2/3 \end{bmatrix} \begin{bmatrix} 2 & 1 & 1 \\ 0 & 3 & 8 \\ 0 & 0 & -2/3 \end{bmatrix}$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 3 & 8 \\ 0 & 0 & -2/3 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 1 & 1 \\ 0 & 3 & 8 \\ 0 & 0 & -2/3 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$26) a) S = \begin{bmatrix} 1 & 3 & 0 \\ 3 & 11 & 4 \\ 0 & 4 & 9 \end{bmatrix}$$

$$E_{21} = \begin{bmatrix} 1 & 3 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad E_{21}^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$E_{21} S E_{21}^T = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 0 \\ 3 & 11 & 4 \\ 0 & 4 & 9 \end{bmatrix} \begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 4 \\ 0 & 4 & 9 \end{bmatrix} \begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 4 \\ 0 & 4 & 9 \end{bmatrix}$$

$$b) E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix} \quad E_{32}^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$E_{32} E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 6 & -2 & 1 \end{bmatrix}$$

$$E_{21}^T E_{32}^T = \begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -3 & 6 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ -6 & -2 & 1 \end{bmatrix} \quad L^T = \begin{bmatrix} 1 & 3 & -6 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$29) a) A = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 1 & 0 & -1 \end{bmatrix} \quad A^T = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \\ 0 & -1 & -1 \end{bmatrix}$$

$$A^T y = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \\ 0 & -1 & -1 \end{bmatrix} \begin{bmatrix} y_{BC} \\ y_{CS} \\ y_{BS} \end{bmatrix} \quad [3 \times 3][3 \times 1]$$

$$= \begin{bmatrix} y_{BC} + y_{BS} \\ y_{CS} - y_{BC} \\ -(y_{CS} + y_{BS}) \end{bmatrix} \quad \checkmark$$

b) Either way: $(Ax)^T y = x^T (A^T y)$

$$x^T = [x_B \quad x_C \quad x_S]$$

$$0_0 = x^T (A^T y) = [x_B \quad x_C \quad x_S] \begin{bmatrix} y_{BC} + y_{BS} \\ -y_{BC} + y_{CS} \\ -y_{CS} - y_{BS} \end{bmatrix}$$

$$= x_B y_{BC} + x_B y_{BS} - x_C y_{BC} + x_C y_{CS} - x_S y_{CS} - x_S y_{BS}$$

$$\Rightarrow 6 \text{ terms} \quad \checkmark$$

31) $Ax \cdot y \rightarrow \text{cost of } \Phi \quad \checkmark$

$x \cdot A^T y \rightarrow \text{outputs} \quad \checkmark$

33)

• Problem Set 3.1 - Vector Spaces

1) $(x_1, x_2) + (y_1, y_2) \rightarrow (x_1 + y_2, x_2 + y_1)$
 $c\mathbf{x} = (cx_1, cx_2)$

$\Rightarrow$ Condition 7 is not satisfied.

$\hookrightarrow \mathbf{x} + \mathbf{y} \neq \mathbf{y} + \mathbf{x}$

$\hookrightarrow \mathbf{x} + (\mathbf{y} + \mathbf{z}) \neq (\mathbf{x} + \mathbf{y}) + \mathbf{z}$

$\hookrightarrow (c_1 + c_2)\mathbf{x} \neq c_1\mathbf{x} + c_2\mathbf{x}$

3) a). $\mathbf{x} > 0$ in $\mathbb{R}'$

$\hookrightarrow$ Rules broken:

$\hookrightarrow$ No $\mathbf{0}$ (4)

$\hookrightarrow$ No $-\mathbf{x}$ (5)

$\hookrightarrow c\mathbf{x}$ may not be in the set. (6)

b) $\mathbf{x} + \mathbf{y}$; $c\mathbf{x} \Rightarrow \mathbf{x}\mathbf{y}$; $\mathbf{x}^c$
 $\{c=3, x=2, y=1\}$

$\mathbf{x} + \mathbf{y} = (2) + (1) = 3$ $\mathbf{x}^c = (2)^{(3)} = 8$

~~$c\mathbf{x} = (3)(2) = 6$~~

$y = 0$ acts a zero vector.

4) $A = \begin{bmatrix} 2 & -2 \\ 2 & -2 \end{bmatrix}$; zero vector $\Rightarrow \mathbf{0} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ ✓
 $\frac{1}{2}A = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$ ✓
 $-A = \begin{bmatrix} -2 & +2 \\ -2 & +2 \end{bmatrix}$ ✓

What matrices are in the smallest Subspace containing A ?

$\hookrightarrow cA$

$$6) \quad f(x) = x^2 \quad -\infty < x < \infty$$

$$g(x) = 5x$$

$$h(x) = 3f(x) - 4g(x)$$

$$= 3(x^2) - 4(5x)$$

$$= 3x^2 - 20x \quad \checkmark$$

$$8) \quad f(x) \text{ and } g(x) \quad f(g(x))$$

↳ zero vector would be $g(x) = x$.

$$cf(x)$$

↳ two rules that are broken:

$$1. \quad x + y \neq y + x$$

$$f(g(x)) \neq g(f(x)) \quad \checkmark$$

~~1. unique vector x for $x + (-x) = 0$~~

$$2. \quad (c_1 c_2)x \neq c_1(c_2 x)$$

$$(f + g)(x) = f(g(x))$$

$$\text{then } (g + f)(x) = g(f(x))$$

↗ ≠

→ Rule 4 broken

$$\hookrightarrow f^{-1}(x)$$

$$\rightarrow f(f^{-1}(x)) = x$$

9) a) $\begin{cases} x+y \rightarrow \text{same set} \\ cx \rightarrow \text{not in same set} \end{cases} \left\{ \text{all vectors in } \mathbb{R}^2 \text{ with } \underline{\text{integer}} \text{ components} \right\}$

b) union of two lines through the origin

11) $\mathbb{C}^{1 \times p}$

a) $CA \quad A = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}; \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \right\} \times \begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix}$

b) $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} a & a \\ 0 & 0 \end{bmatrix} \checkmark$

c) $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \& \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \checkmark$

~~11)~~

19) $A = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} \rightarrow \text{Line through origin in } \mathbb{R}^3 \checkmark (x, 0, 0)$

$B = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 0 & 0 \end{bmatrix} \rightarrow \text{Plane through origin in } \mathbb{R}^3 (x, y, 0)$

$C = \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 0 & 0 \end{bmatrix} \rightarrow \text{Point / Arrow Vector Vector through origin in } \mathbb{R}^3$
 $\hookrightarrow \text{line of Vectors} \quad (x, 2x, 0)$

21) $\rightarrow$ No idea?

$\rightarrow$ Which two matrices have same column? A and C

$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} \quad C \text{ is combination of columns}$
 $\hookrightarrow \text{Scalable of } A$

$B^T = C \times \hookrightarrow \text{Occupy same column space}$

23) Add extra column b to matrix A

↳ Column space gets larger unless new column is dependent on an existing column $\rightarrow \checkmark$ if column added is already inside A 's column space.

$$A = \begin{bmatrix} 2 & 1 \\ 3 & 6 \end{bmatrix}$$

$$A_{\text{larger}} = \begin{bmatrix} 2 & 1 & 5 \\ 3 & 6 & 7 \end{bmatrix}$$

$$A_{\text{Not larger}} = \begin{bmatrix} 2 & 1 & 4 \\ 3 & 6 & 6 \end{bmatrix} \quad \begin{matrix} 2 \\ 3 \end{matrix} \begin{matrix} \times 2 \\ \times 2 \end{matrix} \rightarrow \begin{matrix} 4 \\ 6 \end{matrix}$$

Why $Ax = b$ solvable doesn't get larger?

↳ No new unknowns are added.

↳

$$[A \ b] = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \rightarrow \text{larger col space}$$

↳ no sol to $Ax = b$

$$[A \ b] = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \rightarrow b \text{ in column space}$$

↳ $Ax = b$ has solution.

25) $\left. \begin{matrix} Ax = b \\ Ay = b^* \end{matrix} \right\} \text{solvable}$ then: $Az = b + b^*$ is ~~is~~ solvable. What is z ?

$$b + b^* = Az$$

$$= A(x + y)$$

$$\therefore z = x + y$$

$$b + b^* = Az$$

$$Ax + Ay = Az$$

$$x + y = z$$

$$\therefore z = x + y \quad \checkmark$$

27)

a) False $\checkmark \rightarrow$ Vectors not in column space cannot be a subspace with a space they are outside of.

b) True $\checkmark$

c) True $\checkmark$

d) $A = \begin{bmatrix} 2 & 4 \\ 3 & 1 \end{bmatrix}$ $A - I = \begin{bmatrix} 2 & 4 \\ 3 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 4 \\ 3 & 0 \end{bmatrix}$
 $\hookrightarrow$ equal to A's column space?

col space $C(A) \neq C(A - I)$
 $\text{col}\left(\begin{bmatrix} 2 & 4 \\ 3 & 1 \end{bmatrix}\right) \neq \text{col}\left(\begin{bmatrix} 1 & 4 \\ 3 & 0 \end{bmatrix}\right)$

$\therefore$ False $\checkmark$

29) 9×12 system: $Ax = b$ solvable for every b
 then $C(A) = C(b)$. $\times$
 $\hookrightarrow$ every b in $C(A)$ $\checkmark$

So that space is: $\mathbb{R}^9$

$9 \begin{bmatrix} 1 & \dots & 12 \\ \vdots & & \\ 9 \end{bmatrix}$ each column exist in $\mathbb{R}^9$ space.

$C(A) = \mathbb{R}^9$

$C(b) \rightarrow$ space of single vector

$\hookrightarrow b$ in $\mathbb{R}^9$